

Solução do Problema 25fv47 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: 25fv47
- Número de Variáveis: 1572
- Número de Restrições: 821
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 0.000e+00
- Valor Primal: 5.191e+03
- Valor Dual: 5.191e+03
- Gap: 6.412e-19
- Número de Iterações: 27

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z) com valores ≥ 0 do problema 25fv47

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	5.778e+01	2	1.000e-03
4	2.798e+01	3	1.000e-03
6	7.350e+00	5	1.000e-03
8	1.500e+01	7	1.000e-03
9	3.191e+02	14	5.000e-02
11	3.860e+02	58	9.822e+01
13	2.164e+03	60	1.551e+01
15	6.250e+01	61	5.406e+00
16	1.136e+03	62	4.028e+00
18	4.242e+02	63	4.774e+00
19	8.180e+01	64	4.153e+00
20	1.555e+02	65	1.395e+01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
21	1.171e+02	66	2.004e+00
22	2.288e+01	67	3.671e+00
23	8.861e+01	68	2.351e+00
25	1.908e+02	69	2.948e+00
26	2.809e+00	70	3.406e+00
27	3.579e+02	71	4.449e+00
28	1.836e+02	82	4.258e-01
29	1.047e+02	83	5.965e-01
30	3.171e+02	84	3.913e-01
31	1.526e+02	85	7.947e-01
32	2.124e+02	86	5.386e-01
33	7.102e+01	87	3.621e-01
34	2.361e+02	88	6.518e-01
36	3.591e+01	89	4.664e-01
37	1.510e+02	90	6.577e-01
38	1.260e+02	91	2.095e-01
39	1.422e+01	93	1.732e-01
40	7.866e+02	94	1.122e-01
41	2.049e+01	95	3.023e-02
42	3.769e+01	96	3.471e-01
43	8.454e+01	97	1.759e-01
44	1.155e+02	99	2.722e-01
48	7.900e+01	100	1.316e-01
49	1.480e+02	101	3.706e-01
51	1.103e+02	102	3.845e-02
52	3.569e+01	103	2.343e-01
56	5.209e+01	104	1.049e-01
57	1.949e+02	106	3.328e-01
59	3.641e+01	107	1.650e-01
72	1.021e+01	109	2.514e-01
74	1.538e+02	110	1.228e-01
75	1.497e+02	111	2.362e+00
76	1.089e+02	112	2.330e+00
77	2.777e+00	113	2.364e+00
78	3.383e+01	115	3.703e-01
79	1.714e+01	116	1.826e+00
80	2.443e+00	117	1.794e+00
81	9.329e-01	118	1.827e+00
92	2.543e+01	119	1.850e+00
98	3.377e+00	123	1.958e-02
105	2.628e+01	128	1.239e-02
108	4.173e+01	129	2.219e-01
114	1.393e+01	130	2.219e-01
120	7.000e+00	132	1.139e-02
121	7.042e+00	134	2.229e-01
122	1.007e+01	135	2.229e-01
124	3.115e-01	140	3.001e-01
125	9.072e+00	141	3.001e-01
126	2.409e+01	143	5.280e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
127	7.322e-01	144	6.882e-02
131	9.779e+00	145	5.225e-02
133	4.474e+00	146	3.334e-01
136	7.479e+00	147	3.334e-01
137	5.969e+00	149	5.241e-01
138	5.791e+00	150	5.241e-01
139	3.318e+00	151	8.716e-02
142	8.982e+00	152	5.432e-01
148	6.122e+01	155	9.484e-01
159	2.965e+00	156	1.017e+00
160	1.326e+00	157	9.484e-01
162	5.688e+00	158	1.017e+00
163	1.392e+01	161	9.983e-02
168	4.500e+01	164	1.054e-01
169	4.640e+01	165	1.270e-01
171	6.173e+00	166	3.891e-01
172	6.947e+00	167	3.891e-01
174	8.173e+01	170	5.133e-02
175	5.902e-01	173	1.147e+00
177	2.396e+01	176	1.805e-01
179	2.211e+01	178	1.805e-01
189	2.374e+01	180	1.459e-01
191	4.750e+00	181	1.236e+00
197	1.455e+00	182	1.222e+00
199	2.048e+01	183	1.224e+00
200	9.017e-01	184	1.219e+00
201	1.220e+01	185	1.236e+00
202	1.396e+01	186	1.222e+00
203	1.732e+01	187	1.224e+00
204	8.403e+00	188	1.219e+00
207	5.000e+00	190	4.616e-01
209	1.308e+01	192	1.045e+00
210	4.932e+01	193	4.587e-01
211	5.000e+00	194	1.409e+00
213	2.000e+00	195	4.431e+01
214	5.792e+00	206	4.759e-02
216	2.493e+00	212	9.473e-02
219	3.868e+01	215	2.386e+00
231	2.396e+01	217	9.463e-01
233	8.000e+01	218	9.909e+00
234	2.800e+01	220	9.996e-03
235	4.694e+00	221	2.271e-02
236	8.873e+01	223	4.569e-01
239	2.707e+01	224	4.569e-01
241	1.511e+02	225	5.562e-02
242	2.491e+02	226	1.680e-01
243	1.890e+02	227	5.562e-02
244	3.685e+01	228	1.680e-01
245	2.557e+02	229	7.400e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
247	7.969e+00	232	6.122e-01
249	2.689e+00	237	3.512e-02
250	4.360e+01	238	7.734e-02
256	1.985e+01	240	3.858e-01
260	2.777e+01	246	2.922e-03
262	1.981e+01	248	5.861e-03
263	1.100e+00	252	3.203e-02
264	2.290e+00	253	9.827e+01
266	3.344e+00	254	5.770e+00
268	5.647e+00	257	3.661e-02
269	2.337e+00	258	5.270e+00
271	2.625e+01	259	3.346e-01
275	3.832e+01	261	3.203e-01
282	5.000e+00	265	5.710e-01
290	2.115e+01	267	6.712e-03
293	1.648e+01	270	5.787e+00
296	2.772e+01	272	1.178e+00
298	3.134e+01	273	4.230e-01
302	2.121e+01	274	8.355e-02
303	1.916e+01	276	8.003e-01
305	6.023e+00	277	3.996e-01
308	4.634e+01	278	2.603e-01
317	1.443e+01	279	1.000e+00
318	1.335e+01	280	8.821e-01
319	8.086e-01	281	5.599e-03
320	1.343e+01	283	4.975e+00
325	2.073e+00	284	4.836e+00
326	3.297e+00	285	4.660e+00
327	4.501e+01	286	1.137e-01
329	1.726e+01	287	6.143e-02
334	2.399e+01	288	1.094e-01
335	6.076e+00	289	3.563e-01
336	5.981e+01	291	3.759e-01
337	1.526e-02	292	1.714e-01
343	2.662e+01	294	2.339e-02
345	1.678e+00	295	4.378e-03
349	9.309e+00	297	8.347e-02
350	2.130e+00	299	4.862e-01
355	1.785e+01	300	4.140e-01
358	1.990e+00	301	3.461e-01
360	7.209e+00	304	3.772e-01
361	1.139e+00	306	2.285e-03
371	1.348e+02	307	5.364e-01
374	3.896e+01	309	9.267e-01
377	1.016e+01	310	3.601e-02
378	1.353e+01	311	3.601e-02
379	6.163e+00	312	3.436e-02
382	4.460e+01	313	4.958e-01
383	1.201e+02	314	1.914e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
385	1.277e+01	315	8.116e-01
389	4.701e+00	316	6.029e-01
390	1.030e+01	321	7.588e-02
391	1.802e+01	322	1.896e+00
392	1.477e+01	323	8.017e-01
396	6.056e+01	324	7.238e-01
400	1.197e+01	328	4.856e-03
403	4.553e+01	330	1.895e+00
411	1.295e+01	331	8.001e-01
412	6.974e+01	332	9.492e-01
414	2.800e+01	333	3.785e-03
416	4.698e+01	338	8.389e-02
422	6.013e+00	339	9.608e-01
424	6.811e-01	340	9.553e-01
426	3.528e+00	341	1.384e+00
431	1.037e+01	342	3.703e-03
432	1.776e+01	344	6.613e-02
436	2.012e+00	347	1.489e-02
438	2.018e+01	348	5.127e-01
447	1.400e+01	351	2.478e+00
449	1.517e+01	352	1.780e+00
450	3.413e+00	353	1.455e+00
453	2.500e+01	354	1.170e-02
454	6.091e+01	356	6.850e-01
455	3.100e+01	357	7.334e-01
458	2.181e+01	359	7.004e-02
459	3.166e+01	362	2.349e-01
461	4.900e+01	363	2.545e-01
463	1.800e+00	364	3.795e-01
464	3.300e+01	365	5.446e-01
470	6.000e+01	366	4.792e-01
472	3.288e+02	367	3.597e-01
474	2.110e+02	368	5.898e-01
476	1.408e+01	369	6.656e-03
477	2.265e+01	370	5.074e-01
478	1.817e+02	372	9.234e-02
479	4.800e+00	373	8.860e-02
480	1.794e+00	375	5.492e-03
482	2.359e+00	376	9.942e-02
483	2.093e+00	380	6.328e-02
484	3.004e+00	381	6.826e-02
485	1.406e+00	384	2.134e-02
486	7.650e+00	386	1.095e-01
487	2.986e+00	387	9.237e-02
494	1.660e+00	388	2.535e-01
496	1.752e+01	393	2.131e-01
500	1.304e+01	394	3.197e-01
504	5.559e+00	395	1.424e-01
510	3.512e+01	397	1.541e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
517	1.120e+01	398	5.171e-01
518	1.251e+01	399	3.580e-01
521	1.647e+00	401	4.912e-01
523	1.367e+01	402	2.096e-01
525	4.311e+00	404	2.047e+00
527	5.209e-01	405	2.099e+00
529	4.128e+00	406	1.541e-01
530	1.499e+00	407	1.803e+00
531	8.479e+00	408	1.880e+00
532	7.672e+00	409	3.576e-01
537	7.007e+01	410	4.909e-01
538	2.262e+01	413	2.942e-01
544	9.240e+00	415	3.319e-01
545	4.082e+00	417	3.795e-01
546	1.583e+01	418	4.591e-01
547	6.963e+00	420	9.417e-02
548	5.600e+01	421	3.612e-01
549	1.700e+01	423	3.744e-01
550	1.100e+01	425	5.040e-01
551	1.800e+01	427	1.090e+00
552	5.000e+00	428	4.820e-02
553	7.200e+01	429	1.647e-02
554	1.292e+01	430	1.595e-02
555	2.300e+01	433	1.256e+00
556	1.700e+01	434	1.492e+00
559	2.863e+02	435	6.364e-04
561	4.925e+02	437	5.446e-01
567	1.930e+02	439	1.004e-03
572	6.445e-01	440	6.703e+00
573	3.000e+01	441	1.240e+00
574	5.359e+00	442	1.294e+00
575	4.036e+00	443	2.482e+00
576	2.783e+00	444	2.519e+00
577	3.240e+00	445	1.846e+00
579	1.549e+01	446	1.901e+00
584	4.858e+00	448	1.251e-02
591	7.583e+00	451	1.897e-01
592	4.508e+00	452	1.197e-01
594	1.201e+01	456	2.284e+00
595	1.073e+01	457	2.285e+00
597	1.645e+00	460	9.557e-02
598	5.589e+00	462	1.387e-01
599	1.418e+01	465	2.257e+00
600	1.335e+01	466	1.675e+00
602	9.529e+00	467	4.494e+00
603	1.850e+01	468	4.507e+00
606	1.691e+01	469	4.151e+00
607	1.517e+01	471	2.055e-02
609	1.534e+01	473	1.186e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
610	2.275e+01	475	1.275e+00
611	4.623e+01	481	3.397e-01
616	5.400e+01	489	2.431e+00
625	4.083e+01	490	1.038e-01
628	1.568e+01	491	3.122e-01
631	4.050e+01	492	4.213e-01
635	3.153e+00	493	4.450e-01
636	4.998e+01	495	7.426e-01
639	5.658e+01	497	2.065e-01
647	3.176e+01	498	3.216e-01
652	1.321e+01	499	4.105e-01
655	1.620e+01	501	7.054e-01
657	6.000e+01	502	1.607e+00
658	1.057e+01	503	3.109e-01
668	6.365e+01	505	2.630e-01
671	1.882e+00	506	4.221e-01
672	2.449e+01	507	3.998e-01
673	5.907e+00	508	7.004e-02
674	3.948e+01	509	6.954e-01
677	7.929e+00	511	2.135e-01
678	6.449e+01	512	3.014e-01
681	3.743e+01	513	5.626e-01
682	1.019e+00	514	3.066e-01
683	6.250e+00	515	7.695e-01
691	2.446e+01	516	7.270e-01
692	1.927e+01	519	1.897e-02
697	1.098e+02	520	1.426e-01
698	1.347e+01	522	1.426e-01
700	7.229e+01	524	1.676e-02
701	3.800e+01	526	1.911e-01
707	3.385e+01	528	1.911e-01
710	4.500e+01	533	1.462e-01
711	7.500e+01	534	7.723e-01
714	1.332e+01	535	2.342e-01
715	1.215e+01	536	1.676e-01
721	3.150e+01	539	1.644e+00
722	9.511e+00	540	1.667e+00
723	8.000e+00	541	1.671e+00
724	4.609e+01	542	1.700e+00
725	7.961e+01	543	3.293e-01
726	5.000e+01	557	2.013e-03
728	1.183e+01	558	2.994e-01
736	3.100e+01	560	2.492e-02
743	1.215e+01	562	1.695e-02
745	6.315e+01	563	1.732e+00
751	5.548e+01	564	2.466e+00
752	3.114e+02	565	6.175e-02
757	2.494e+01	570	6.523e-02
759	6.253e+01	571	6.302e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
760	6.085e+01	578	2.754e+00
761	1.784e+00	580	4.097e+01
765	4.273e+01	581	2.374e-01
767	5.773e+01	582	4.970e-01
768	3.313e+00	583	9.584e-01
770	4.697e+01	585	1.300e+00
771	3.264e+01	586	2.758e-01
775	1.131e+00	587	7.024e+00
776	7.256e+00	588	3.376e-01
783	1.451e+01	589	4.190e-01
806	1.238e+01	590	9.639e-01
807	3.865e+01	593	6.125e+00
808	2.447e+01	596	3.212e-01
809	1.713e+01	601	2.361e-01
811	5.331e+01	604	1.577e+01
816	8.773e+01	605	1.603e+01
818	2.103e+02	608	2.022e-02
819	1.536e+02	613	2.555e-01
820	1.200e+01	614	2.563e-01
822	2.630e+01	615	1.025e-01
823	3.414e+02	617	1.872e-01
824	1.600e+01	618	3.526e-01
825	3.556e+01	619	1.320e-01
826	1.936e+02	620	1.509e-01
828	1.036e+02	621	2.223e-01
830	1.465e+02	622	2.462e-02
831	7.032e+01	623	9.988e-02
832	7.822e+01	624	2.291e-01
836	1.329e+00	626	1.374e-01
838	1.200e+00	627	1.041e-01
840	1.322e-01	629	1.512e-02
845	4.049e+00	630	4.382e-02
846	1.875e+00	632	1.571e-02
848	1.685e+01	633	1.226e-02
851	2.351e+01	634	7.096e-02
852	1.400e+01	637	5.925e-02
853	1.000e+01	638	1.648e-02
854	2.139e+01	640	1.776e-01
855	1.857e+01	641	2.449e-02
856	1.333e+00	642	6.307e-02
861	1.955e+01	643	1.581e+00
867	1.572e+01	644	8.279e-02
869	1.650e+01	645	1.691e-01
875	1.545e+01	646	3.026e-01
876	2.441e+01	648	1.015e-01
877	4.923e+00	649	2.376e-01
878	1.234e+01	650	6.538e-01
882	1.316e+01	651	1.513e+00
884	1.453e+01	653	1.052e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
885	2.315e+01	654	1.094e+00
886	2.081e+01	656	2.771e-01
898	1.728e+01	659	1.784e-01
904	1.435e+00	660	4.177e-01
905	2.108e+00	661	3.924e-01
906	5.543e+00	662	4.017e-01
909	4.434e+00	663	4.023e-01
913	1.965e+00	664	3.660e-01
915	1.590e+00	665	3.640e-01
921	1.554e+01	666	3.688e-01
923	1.059e+02	667	3.688e-01
932	2.403e+01	669	2.000e-02
933	2.882e+01	670	3.125e-02
934	5.349e+01	675	1.348e-01
937	2.799e+01	676	9.184e-02
938	3.410e+01	679	2.967e-02
945	2.008e+01	680	4.677e-02
947	5.862e+01	684	4.807e-02
948	8.161e+00	685	5.493e-02
949	1.760e-01	686	2.026e-01
951	4.286e+01	687	2.202e-01
957	1.952e+01	688	1.275e+00
958	4.591e+01	689	6.679e-01
966	9.430e+00	690	1.296e+00
967	9.130e+00	693	5.057e-01
968	2.335e+01	694	1.188e+00
970	1.000e+01	695	3.480e-01
971	5.366e+01	696	2.688e-01
973	1.481e+01	699	4.192e-01
976	1.716e+01	702	4.845e+00
977	6.867e+00	703	4.503e+00
978	4.205e+00	704	4.463e+00
979	6.265e-01	705	4.672e+00
980	1.968e+00	706	5.046e+00
981	1.398e+00	708	3.908e-01
984	7.620e+00	709	7.740e+00
989	1.758e+01	712	3.045e-01
990	9.650e+00	713	1.836e-02
993	1.979e+01	716	2.943e-02
994	2.208e+00	717	8.698e-02
995	1.100e+01	718	8.466e-02
998	6.000e+00	719	7.192e-02
999	5.200e+01	720	1.000e-03
1000	2.780e+01	729	4.923e+00
1001	3.600e+01	730	1.143e+01
1005	5.000e+01	731	3.776e-01
1006	1.000e+00	732	1.446e-01
1007	8.000e+00	733	3.823e+00
1008	6.776e+00	734	3.941e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1009	2.646e+01	735	1.159e+01
1010	4.239e+01	737	1.885e-03
1012	5.988e+01	738	5.331e-02
1014	2.353e+02	739	5.331e-02
1016	1.889e+02	740	6.759e+00
1017	2.376e+01	741	6.889e+00
1018	3.533e+01	742	4.681e+00
1019	3.139e+02	744	4.781e+00
1020	3.850e-02	746	4.099e+00
1021	8.099e+01	747	1.242e-01
1022	1.016e+01	748	4.102e+00
1023	2.774e+00	749	3.590e-01
1024	1.865e+00	750	3.610e-01
1025	8.474e-01	753	3.845e-01
1026	4.690e+00	754	3.845e-01
1028	2.616e-01	755	3.328e-01
1029	1.640e+00	756	3.331e-01
1030	9.182e-01	758	5.426e-04
1031	1.821e+01	762	4.757e-04
1037	3.574e+01	764	1.199e-03
1038	6.256e+00	766	2.177e-03
1050	4.927e+00	769	7.045e-04
1051	2.838e+01	772	5.664e-01
1053	9.761e+00	773	5.672e-01
1055	4.084e+00	774	3.579e-01
1061	1.749e+01	777	1.702e-01
1065	2.055e+01	778	1.730e+00
1067	1.700e+01	779	1.657e-01
1069	3.994e+01	780	2.259e-03
1071	1.776e+01	782	2.396e-03
1077	5.571e+00	784	3.447e-03
1083	1.495e+01	787	3.420e-01
1084	2.660e+00	788	4.171e-02
1085	2.516e+01	789	4.142e-02
1090	2.220e+01	791	3.310e-02
1091	3.433e+01	792	6.439e-01
1095	2.001e+01	793	6.446e-01
1101	6.218e+00	795	4.976e-01
1107	7.989e+00	796	3.359e-02
1109	6.718e+00	797	3.378e-02
1114	3.313e+00	799	3.065e-01
1116	9.792e+01	800	8.810e-01
1117	1.980e+01	801	2.107e-01
1120	8.748e+01	802	4.826e-01
1122	1.136e+01	803	4.813e-01
1125	1.254e+01	804	5.302e-01
1128	1.601e+02	805	5.278e-01
1132	3.385e+01	810	1.297e-03
1137	1.700e+01	812	1.847e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1139	1.524e+01	813	8.330e-02
1141	1.000e+00	814	3.025e-02
1143	2.015e+00	815	6.497e-02
1145	8.000e+00	817	2.308e-03
1148	1.930e+02	821	7.526e-01
1149	4.000e+00	833	2.972e-02
1164	2.857e+00	834	3.764e-02
1168	1.643e+00	837	6.395e-01
1169	2.857e+00	839	1.370e+01
1171	4.500e+00	841	6.048e+00
1174	1.600e+01	842	6.395e-01
1179	9.056e-01	843	7.449e+00
1181	2.015e+00	844	2.188e+00
1191	7.700e+01	849	8.197e-01
1193	7.143e-01	850	3.481e-01
1197	2.465e+01	857	1.762e-01
1198	1.935e+01	858	1.692e-01
1200	1.207e+01	859	4.158e-01
1202	1.493e+01	860	1.624e+00
1203	2.243e+01	862	1.811e-01
1214	2.506e+01	863	1.761e-01
1217	2.360e-01	864	4.158e-01
1218	1.821e+01	865	1.624e+00
1220	3.942e+00	866	3.972e-02
1229	3.767e+00	868	2.566e-02
1232	1.623e+01	870	3.555e-02
1233	1.400e+01	871	2.745e-02
1240	1.100e+01	872	2.666e-02
1242	4.400e+01	873	8.345e-01
1244	1.800e+01	874	1.866e+00
1245	1.600e+01	879	9.141e-01
1248	4.110e+00	880	4.014e+00
1249	1.100e+01	881	1.912e+00
1250	9.540e+00	883	4.956e-02
1251	3.035e+01	887	1.185e-01
1258	2.300e+01	888	1.167e-01
1261	1.707e+01	889	2.072e-01
1266	1.400e+01	890	2.063e+00
1269	1.900e+01	891	3.758e+00
1274	2.400e+01	892	3.998e-02
1275	4.000e+00	893	8.454e-02
1280	8.113e+01	894	8.712e-02
1281	1.487e+02	895	5.072e-01
1282	3.418e+01	896	1.319e-01
1285	1.000e+02	899	9.772e-02
1287	1.380e+01	900	7.616e-03
1296	1.517e+01	901	1.434e+00

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores λ 0 do problema
25fv47

Coordenada y	Valor y (Dual)
2	1.962e+00
3	1.926e+00
4	1.962e+00
5	9.625e+00
6	8.784e+00
7	8.333e+00
8	7.665e+00
9	8.120e+00
10	6.622e+00
11	6.622e+00
12	6.583e+00
13	6.297e+00
14	5.932e+00
15	6.264e+00
16	3.755e+00
17	3.755e+00
18	1.701e+00
19	1.701e+00
20	1.483e+00
21	1.430e+00
22	1.329e+00
23	1.256e+00
26	1.000e-01
27	7.600e-02
28	1.010e-01
29	8.850e-02
32	6.374e-01
39	2.587e-01
41	3.525e+00
42	3.479e+00
43	3.525e+00
44	2.988e+00
45	2.988e+00
46	1.139e+00
47	1.032e+00
48	1.028e+00
49	8.307e-01
50	8.307e-01
51	3.967e-01
52	3.241e-01
53	9.875e-01
54	9.766e-01
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Coordenada y	Valor y (Dual)
55	1.797e+00
56	2.197e+00
57	9.280e-01
58	1.092e+00
61	8.715e-01
62	8.636e-01
63	4.153e+00
64	5.041e+00
65	7.680e-01
66	5.056e-01
67	5.056e-01
68	1.652e+00
73	4.490e+00
74	3.542e+00
75	1.904e+00
76	1.036e+00
77	1.973e+00
78	1.054e+00
80	7.653e-01
81	9.097e-01
83	1.582e-02
84	4.875e-01
85	1.582e-02
88	4.976e-01
89	3.259e+00
137	4.433e+00
138	3.642e+00
148	1.730e+00
149	1.730e+00
150	1.730e+00
151	1.730e+00
152	7.442e+00
153	7.442e+00
154	7.108e+00
155	6.630e+00
156	6.491e+00
157	6.300e+00
158	7.196e+00
159	6.278e+00
160	6.120e+00
161	6.119e+00
162	6.117e+00
163	5.759e+00
164	5.843e+00
165	6.615e+00
166	5.734e+00
167	5.978e+00
168	5.647e+00
169	3.851e+00
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Coordenada y	Valor y (Dual)
170	3.849e+00
171	1.603e+00
172	1.515e+00
173	1.361e+00
174	1.493e+00
175	1.253e+00
176	4.218e+00
177	3.972e+00
178	3.966e+00
179	1.001e+00
180	8.112e+00
181	8.706e+00
182	1.435e+00
183	1.894e+00
184	1.814e+00
185	8.364e+00
186	7.296e+00
187	9.754e+00
188	7.057e+00
189	5.567e+00
190	5.561e+00
191	5.561e+00
192	1.491e+00
193	1.730e+00
194	8.740e+00
195	8.488e+00
196	8.613e+00
197	7.403e+00
198	7.766e+00
199	6.871e+00
200	6.871e+00
201	6.518e+00
202	6.675e+00
203	1.130e+01
204	6.318e+00
205	7.623e+00
206	6.341e+00
207	1.111e+01
208	4.700e+00
209	5.968e+00
210	5.855e+00
211	5.647e+00
213	5.732e+00
214	5.636e+00
215	5.303e+00
216	4.094e+00
218	3.614e+00
219	3.633e+00
220	1.307e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
221	1.281e+00
222	1.307e+00
223	4.498e-01
224	1.321e+00
225	1.285e+00
226	1.398e+00
227	1.435e+00
284	7.472e+00
285	9.243e-01
286	1.782e+00
287	9.097e+00
288	6.915e+00
289	7.980e+00
290	5.952e+00
291	9.990e-01
296	3.397e-01
297	3.397e-01
298	3.397e-01
299	3.397e-01
300	2.748e+00
301	4.355e+00
302	4.666e+00
303	5.100e+00
304	5.827e+00
305	1.167e+00
306	9.186e-01
307	9.354e-01
308	7.395e-01
309	6.925e-01
310	4.896e-01
311	5.186e-01
312	2.751e-01
313	5.423e-01
314	4.578e-01
315	5.926e+00
316	5.705e+00
317	4.899e+00
318	2.748e+00
319	4.664e+00
320	3.009e+00
321	8.253e-01
322	8.253e-01
323	1.091e+00
324	2.854e-01
325	4.713e-01
326	4.418e-01
327	5.801e-01
328	3.890e-01
330	1.474e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
331	5.787e-01
332	2.854e-01
362	4.671e-01
374	4.531e+00
375	4.531e+00
376	1.778e+00
377	4.550e+01
378	6.747e+00
379	1.105e+01
380	8.005e+00
381	9.425e+00
382	2.543e+01
383	7.088e+00
384	6.747e+00
385	1.540e+00
386	1.576e+00
387	1.852e+00
388	8.469e+00
389	6.376e+00
390	6.395e+00
391	6.395e+00
392	6.376e+00
393	6.406e+00
394	6.035e+00
395	6.037e+00
396	6.053e+00
397	1.634e+00
398	2.248e+00
399	1.616e+00
400	1.814e+00
401	1.568e+00
402	6.568e+00
403	6.164e+00
404	5.978e+00
405	9.936e+00
406	5.994e+00
407	6.365e+00
408	1.923e+00
409	1.555e+00
410	1.423e+00
411	1.471e+00
412	1.924e+00
413	1.394e+00
414	6.204e+00
415	6.891e+00
416	1.750e+01
417	1.923e+00
418	1.880e+00
419	1.604e+00
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Coordenada y	Valor y (Dual)
420	1.772e+00
421	1.587e+00
422	2.177e+00
423	1.395e+00
424	2.126e+01
425	1.672e+01
426	1.437e+00
427	1.420e+00
428	9.534e+00
429	9.446e+00
430	8.870e+00
431	9.000e+00
432	7.626e+00
433	8.000e+00
434	1.808e+01
435	9.094e+00
436	1.133e+01
437	6.747e+00
438	6.747e+00
439	7.743e+00
440	7.414e+00
441	7.414e+00
442	6.243e+00
443	6.125e+00
444	5.711e+00
445	5.839e+00
446	6.000e+00
447	5.832e+00
448	3.029e+00
449	3.325e+00
450	3.325e+00
451	1.526e+00
452	1.532e+00
453	1.532e+00
454	7.790e-01
455	1.521e+00
456	1.542e+00
457	1.532e+00
458	1.536e+00
504	7.825e+00
505	1.532e+00
516	2.375e+00
520	1.777e+00
521	1.777e+00
536	2.363e+00
537	3.003e+00
539	3.003e+00
540	4.551e+00
541	6.415e+00

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Coordenada y	Valor y (Dual)
542	6.415e+00
543	6.415e+00
544	6.415e+00
545	6.415e+00
546	6.159e+00
547	5.864e+00
548	5.864e+00
549	5.890e+00
550	5.890e+00
551	5.900e+00
552	5.950e+00
553	5.900e+00
554	5.879e+00
555	3.946e+00
556	4.006e+00
557	4.179e+00
558	1.989e+00
559	1.908e+00
560	2.109e+00
561	1.908e+00
562	1.752e+00
563	2.648e+00
564	2.228e+00
565	2.054e+00
566	2.587e+00
567	3.964e+00
568	3.873e+00
569	1.707e+00
570	1.791e+00
571	1.701e+00
572	1.882e+00
573	2.078e+00
574	1.746e+00
575	7.901e+00
576	9.377e+00
577	5.558e+00
578	5.936e+00
579	3.134e+00
580	3.134e+00
581	1.986e+00
587	9.553e+00
588	8.779e+00
589	8.954e+00
590	7.262e+00
591	7.771e+00
592	6.997e+00
593	6.415e+00
594	7.637e+00
596	7.639e+00
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Coordenada y	Valor y (Dual)
597	6.020e+00
598	5.712e+00
599	5.777e+00
600	5.611e+00
601	5.575e+00
602	5.622e+00
603	5.495e+00
604	3.129e+00
605	3.129e+00
606	2.033e+00
607	2.307e+00
608	1.986e+00
609	1.986e+00
610	9.912e-01
665	1.000e+00
666	1.035e+00
667	5.257e+00
668	5.257e+00
669	6.285e+00
670	6.276e+00
671	5.257e+00
672	5.547e+00
673	5.257e+00
674	6.756e+00
675	6.465e+00
676	3.694e+00
677	2.703e+00
678	3.559e+00
679	2.163e+00
680	3.928e+00
681	2.428e+00
682	1.881e+00
683	8.409e-01
684	1.046e+00
685	1.031e+00
686	1.284e+00
687	7.806e-01
688	1.242e+00
689	8.624e-01
690	7.018e+00
691	6.405e+00
693	5.087e+00
694	7.360e+00
695	5.970e+00
696	8.437e+00
698	6.096e+00
699	6.297e+00
700	4.275e+00
701	4.491e+00
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Coordenada y	Valor y (Dual)
702	7.323e-01
703	3.424e+00
704	8.624e-01
705	8.624e-01
706	1.181e+00
707	1.181e+00
708	2.064e+00
743	7.662e+00
744	6.529e+00
745	7.316e+00
746	7.103e+00
747	5.605e+00
748	1.161e+00
749	9.380e-01
750	1.063e+00
753	7.410e+00
754	7.850e+00
755	7.471e+00
756	7.240e+00
765	5.911e+00
766	8.084e+00
767	7.082e+00
768	1.219e+00
769	1.735e+00
770	1.093e+00
771	8.280e-01
772	8.280e-01
773	2.267e+01
774	1.813e+01
775	7.170e+00
776	6.163e+00
777	6.254e+00
778	1.785e+00
779	6.232e+00
788	1.983e+00
789	1.764e+00
790	1.764e+00
791	3.019e+00
792	4.701e+00
793	1.124e+00
794	1.877e+00
795	7.050e+00
796	3.839e+00